Phase 3, Block 1: The explicit FTCS scheme for the Black–Scholes PDE

Quant Finance Portfolio

1 Setting

This block implements the Forward-Time, Centred-Space (FTCS) finite- difference pricer for European calls and puts. We use the constant-coefficient transformed Black–Scholes equation derived in Block 0,

$$\frac{\partial V}{\partial \tau} = \frac{\sigma^2}{2}\frac{\partial^2 V}{\partial x^2} + \mu\,\frac{\partial V}{\partial x} - r\,V, \qquad \mu = r - \tfrac{\sigma^2}{2}, \tag{1}$$

with $x = \ln(S/K)$, $\tau = T - t$, on the truncated domain $[x_{\min}, x_{\max}] \times [0, T]$ with Dirichlet boundary conditions specified in Block 0. The deliverable is a pricer, validated against the Black–Scholes closed form and analysed in terms of stability and convergence.

2 The FTCS discretisation

Let $V_j^n \approx V(x_j, \tau_n)$ on the uniform grid $x_j = x_{\min} + j\,\Delta x$, $\tau_n = n\,\Delta\tau$. Substituting the centred space stencils and the forward-time stencil into (1):

$$\frac{V_j^{n+1} - V_j^n}{\Delta\tau} = \frac{\sigma^2}{2}\,\frac{V_{j+1}^n - 2\,V_j^n + V_{j-1}^n}{\Delta x^2} + \mu\,\frac{V_{j+1}^n - V_{j-1}^n}{2\,\Delta x} - r\,V_j^n.$$

All spatial differences are evaluated at the *old* time level n. This is what makes the scheme explicit: the new-time value V_j^{n+1} is given directly as a linear combination of three old-time values, without solving any system.

Multiplying by $\Delta\tau$ and isolating V_j^{n+1} on the left,

$$\boxed{V_j^{n+1} \;=\; a_-\,V_{j-1}^n \;+\; a_0\,V_j^n \;+\; a_+\,V_{j+1}^n} \tag{2}$$

with constant coefficients

$$a_- = \alpha - \frac{\mu\,\Delta\tau}{2\,\Delta x}, \quad a_0 = 1 - 2\alpha - r\,\Delta\tau, \quad a_+ = \alpha + \frac{\mu\,\Delta\tau}{2\,\Delta x}, \tag{3}$$

where $\alpha = \frac{\sigma^2}{2}\,\Delta\tau/\Delta x^2$ is the Fourier number from Block 0. The coefficients do not depend on j or n, by virtue of the constant-coefficient structure of (1): this is the structural pay-off of the logarithmic transformation.

3 Probabilistic interpretation

The three coefficients sum to

$$a_- + a_0 + a_+ \;=\; 1 - r\,\Delta\tau.$$

This identity has a clean reading: if a_-, a_0, a_+ are non-negative, define

$$p_- = \frac{a_-}{1 - r\Delta\tau}, \quad p_0 = \frac{a_0}{1 - r\Delta\tau}, \quad p_+ = \frac{a_+}{1 - r\Delta\tau},$$

so that $p_-, p_0, p_+ \geq 0$ and $p_- + p_0 + p_+ = 1$. Then (2) is exactly

$$V_j^{n+1} \;=\; (1 - r\,\Delta\tau)\Big(p_-\,V_{j-1}^n + p_0\,V_j^n + p_+\,V_{j+1}^n\Big) \;\approx\; e^{-r\Delta\tau}\,\mathbb{E}^{\mathbb{Q}}\Big[\,V(x_j + \xi\,\Delta x,\,\tau_n)\,\Big], \tag{4}$$

where $\xi \in \{-1, 0, +1\}$ with $\mathbb{Q}$-probabilities (p_-, p_0, p_+). The factor $1 - r\,\Delta\tau$ is the forward-Euler approximation to the discount factor $e^{-r\Delta\tau}$.

Equation (4) says that FTCS is the discretisation of the risk-neutral pricing identity

$$V(x_j, \tau_{n+1}) \;=\; e^{-r\Delta\tau}\,\mathbb{E}^{\mathbb{Q}}\big[V(X_{n+1}, \tau_n)\,\big|\,X_n = x_j\big]$$

on a discrete state space, with $X_{n+1} - X_n$ taking the values $\pm\Delta x, 0$. This is exactly the structure of a trinomial tree. We will revisit this connection in Block 5: the Kamrad–Ritchken tree is, after a precise reparametrisation, FTCS in disguise.

The reading also highlights why the non-negativity of the coefficients matters: if any $p_\xi < 0$, the operation loses its interpretation as a discrete expectation, and the scheme can amplify oscillations near the strike kink in the payoff.

4 Non-negativity of the coefficients

Three conditions, one per coefficient.

Central coefficient. $a_0 \geq 0$ requires

$$\alpha \;\leq\; \frac{1 - r\,\Delta\tau}{2}. \tag{5}$$

For $r\,\Delta\tau \ll 1$ (always true in this project: $r \leq 0.1$ and $\Delta\tau \leq 10^{-2}$), (5) is essentially the CFL condition $\alpha \leq 1/2$ from Block 0.

Lateral coefficients. $a_\pm \geq 0$ requires $\big|\,\mu\,\Delta\tau/(2\Delta x)\big| \leq \alpha$, equivalently

$$\frac{|\mu|\,\Delta x}{\sigma^2} \;\leq\; 1. \tag{6}$$

The dimensionless ratio $\mathrm{Pe}_h := |\mu|\Delta x/\sigma^2$ is the *grid Péclet number*: it measures the relative weight of convection to diffusion at the scale of one grid cell. For the parameter range of this project ($\sigma \in [0.1, 0.4]$, $|\mu| \leq 0.1$, $\Delta x \leq 0.1$), $\mathrm{Pe}_h \leq 1$ is satisfied with comfortable margin.

Remark 4.1 (Convection-dominated regimes)**.** *When* $\mathrm{Pe}_h > 1$ *(extreme dividends, very small volatility, very coarse grid), centred differences for the convective term produce one negative lateral coefficient, the probabilistic reading breaks, and the scheme exhibits localised oscillations near the strike kink. The standard remedy is an* upwind *discretisation of the convective term: replace* $(V_{j+1} - V_{j-1})/(2\Delta x)$ *by* $(V_j - V_{j-1})/\Delta x$ *if* $\mu > 0$ *or* $(V_{j+1} - V_j)/\Delta x$ *if* $\mu < 0$*. This restores* L^∞ *stability at the price of one order of accuracy in space (*$O(\Delta x)$ *instead of* $O(\Delta x^2)$*). Out of scope for Phase 3.*

Summary. The probability-positive (monotone) regime is

$$\boxed{\alpha \le \tfrac{1}{2} \ \text{ and } \ \mathrm{Pe}_h \le 1.} \tag{7}$$

In this regime the scheme is monotone, and monotonicity implies stability in the uniform L^∞ norm:

$$\max_j |V_j^{n+1}| \ \le \ (1 - r\Delta\tau) \max_j |V_j^n| \ \le \ \max_j |V_j^n|.$$

This is a strictly stronger result than the L^2 stability of the next section, and it is the practical reason the prices computed by FTCS *look smooth*: oscillations cannot grow.

5 Von Neumann stability analysis

The von Neumann analysis of (2) for the full convection-diffusion-reaction equation supplements the previous section with a tight necessary-and-sufficient condition for L^2 stability.

Substituting the Fourier ansatz $V_j^n = g^n e^{ij\theta}$ into (2) and using $e^{i\theta} + e^{-i\theta} = 2 - 4\sin^2(\theta/2)$, $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$, the amplification factor is

$$g(\theta) \ = \ (1 - r\,\Delta\tau) - 4\,\alpha\,\sin^2\tfrac{\theta}{2} + i\,\beta\,\sin\theta, \qquad \beta \ := \ \frac{\mu\,\Delta\tau}{\Delta x}. \tag{8}$$

(β is the unsymmetrised Courant number; it differs from the ν of Block 0 by a factor of 1/2.) The factor is complex: the real part is the diffusive contribution shifted by the reactive term; the imaginary part is the convective contribution.

Proposition 5.1 (Stability of FTCS)**.** *Set $r = 0$ for clarity (the reactive term shifts the analysis trivially and never contributes to instability when $r \ge 0$ and $\Delta\tau \le 1/r$). The condition $|g(\theta)|^2 \le 1$ for all $\theta \in [-\pi, \pi]$ is equivalent to*

$$\alpha \ \le \ \tfrac{1}{2} \qquad \textit{and} \qquad \beta^2 \ \le \ 2\,\alpha. \tag{9}$$

Proof. With $s := \sin^2(\theta/2) \in [0, 1]$, $\sin^2\theta = 4s(1 - s)$. Then

$$|g|^2 = (1 - 4\alpha s)^2 + 4\beta^2 s(1 - s),$$

and after expansion,

$$|g|^2 - 1 = s\left[(16\alpha^2 - 4\beta^2)\,s + (4\beta^2 - 8\alpha)\right].$$

The bracket is linear in s. The condition $|g|^2 \le 1$ for all $s \in (0, 1]$ is equivalent to the bracket being non-positive at both endpoints $s = 0$ and $s = 1$ (linear function on a compact interval attains its extrema at the endpoints). At $s = 0$: $4\beta^2 - 8\alpha \le 0 \Leftrightarrow \beta^2 \le 2\alpha$. At $s = 1$: $16\alpha^2 - 8\alpha \le 0 \Leftrightarrow \alpha \le 1/2$. □ □

The two conditions are not equivalent to monotonicity. The von Neumann conditions (9) are about L^2 stability; the conditions (7) are about monotonicity. The relationship between them is:

- Both conditions share $\alpha \le 1/2$.
- The von Neumann condition $\beta^2 \le 2\alpha$ rewrites as $\mu^2\Delta\tau \le \sigma^2$, a constraint on $\Delta\tau$. For typical parameters ($\sigma = 0.2,\ \mu \le 0.1$): $\Delta\tau \le 4\sigma^2/\mu^2 = 4$ years. Always satisfied.
- The probability-positive condition $\mathrm{Pe}_h \le 1$ rewrites as $|\mu|\Delta x \le \sigma^2$, a constraint on Δx.

The two are different conditions on different variables. The probability-positive constraint is strictly stronger: it implies L^2 stability via monotonicity, but the converse fails. In our parameter range both conditions are satisfied; in convection-dominated regimes, they decouple, and the von Neumann condition can hold without monotonicity (giving stable but oscillatory schemes).

What dominates in practice. The constraint that bites is $\alpha \leq 1/2$. The Péclet condition $\mathrm{Pe}_h \leq 1$ is a constraint on Δx that is comfortably met for the project's parameters. The reactive contribution $r\Delta\tau$ damps every Fourier mode uniformly and never destabilises.

6 The time-marching algorithm

In pseudocode, the inner loop is

```
V[j] for j = 0..N  <-  initial-condition vector at tau = 0
for n = 0, 1, ..., M-1:
    V_new[0] = bc_lower[n+1]                          # Dirichlet
    V_new[N] = bc_upper[n+1]                          # Dirichlet
    for j = 1, ..., N-1:                              # interior nodes
        V_new[j] = a_minus * V[j-1]
                 + a_zero  * V[j]
                 + a_plus  * V[j+1]
    swap(V, V_new)
return V                                              # V at tau = T
```

Three implementation points worth explicit comment.

Double buffer. The update of node j at level $n+1$ uses V_{j-1}^n and V_{j+1}^n. If the implementation overwrites V_j in place, the value of V_j used by the update of V_{j+1} will be the new one rather than the old, breaking the scheme. Two buffers V and V_{new} are needed, swapped after each step.

Boundary conditions are not stencil applications. The boundary nodes $j = 0$ and $j = N$ are fixed by the Dirichlet conditions of Block 0; the stencil is applied only to interior nodes $j = 1, \ldots, N-1$. Trying to apply the stencil at $j = 0$ would require a phantom node $j = -1$ outside the domain.

Recovery of the price at S_0. The grid is in $x = \ln(S/K)$. The pricer must return $V(S_0, 0)$ at the user's spot S_0, which is generally not on a grid node. The V array at $\tau = T$ contains V_j^M for $j = 0, \ldots, N$, and the price is recovered by interpolation at $x_0 = \ln(S_0/K)$. Linear interpolation is enough: it adds an $O(\Delta x^2)$ error consistent with the spatial discretisation order.

7 Convergence and computational cost

Local truncation error. By Taylor expansion of every stencil,

$$\tau_j^n \;=\; O(\Delta\tau) + O(\Delta x^2).$$

First order in $\Delta\tau$ from the forward-Euler step; second order in Δx from the centred space stencils.

Global error and convergence rate. By the Lax equivalence theorem (Block 0, §7), consistency plus stability gives convergence,

$$\left\| V_h - V \right\|_\infty \;=\; O(\Delta\tau) + O(\Delta x^2).$$

Under the CFL constraint $\Delta\tau \leq \Delta x^2/\sigma^2$, both contributions are $O(\Delta x^2)$, so the scheme converges quadratically in Δx. There is no parameter regime in which FTCS achieves better than $O(\Delta x^2)$ convergence: the temporal order is locked at 1, and the spatial order is locked at 2, but the CFL constraint balances the two automatically.

Computational cost. Each time step does $O(N)$ work (one stencil application per interior node). The number of time steps is, by CFL,

$$M \;\geq\; \frac{T\,\sigma^2}{\Delta x^2} \;\sim\; N^2.$$

Total work: $O(N \cdot M) = O(N^3)$. Doubling N multiplies the cost by 8. This N^3 scaling is the price tag of explicit time stepping; Blocks 2 (BTCS) and 3 (Crank–Nicolson) will sidestep it by allowing $\Delta\tau \sim \Delta x$ at the cost of solving a tridiagonal system per step.

8 Implementation outline

The Python module `quantlib.ftcs` exposes `ftcs_european_call` and `ftcs_european_put`, both with signature `(S, K, r, sigma, T, *, N, M, n_sigma=4.0)`. The C++ counterpart in the `quant::pde` namespace has the same arguments. Both validate that $\alpha \leq 1/2$ at construction and raise an error otherwise (Python: `ValueError`; C++: `std::invalid_argument`). The validation is strict and not configurable: the pricer will not silently produce nonsense from a violated CFL.

The validation tests (Python: `validate_ftcs.py`; C++: `test_ftcs.cpp`) check four properties:

1. agreement with the Black–Scholes closed form on a grid of parameters, with tolerance scaled to Δx^2;
2. the empirical convergence rate, by running on (N, M) and $(2N, 4M)$ and checking that the error ratio is close to 4;
3. put-call parity: $C - P = S - Ke^{-rT}$ at every spot, to within twice the discretisation tolerance;
4. that violating CFL ($\alpha > 1/2$) raises the appropriate exception.

A separate diagnostic script, `demo_ftcs_explosion.py`, disables the CFL check and runs the scheme at $\alpha = 0.6$, to visualise the sawtooth instability mode. The output is documented in the manual personal as the Phase 3 numerical lesson.

References

References

[1] D. Tavella and C. Randall. *Pricing Financial Instruments: The Finite Difference Method.* Wiley, 2000. §3.1–3.3 (FTCS for BS, the CFL constraint specific to the equation), §4.2 (convergence).

[2] R. Seydel. *Tools for Computational Finance.* 6th ed., Springer, 2017. §4.2.1 (FTCS), §4.2.4 (stability analysis).

[3] J. C. Strikwerda. *Finite Difference Schemes and Partial Differential Equations.* 2nd ed., SIAM, 2004. §3.4–3.5 (von Neumann analysis for the full CDR equation).

[4] K. W. Morton and D. F. Mayers. *Numerical Solution of Partial Differential Equations.* 2nd ed., CUP, 2005. §2.7–2.10 (Lax equivalence and consistency analysis).

[5] J. Hull. *Options, Futures, and Other Derivatives.* 10th ed., Pearson, 2017. §21.4 (FTCS $\leftrightarrow$ trinomial tree equivalence, from the tree side).

[6] B. Kamrad and P. Ritchken. *Multinomial approximating models for options with k state variables.* Management Science 37(12), 1640–1652, 1991. The original trinomial tree paper; will be revisited in Block 5.